MILITARY SPECIFICATION

FILE, PULP CANAL, DENTAL, HEDSTROM

This specification is mandatory for use by use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers standardized pulp canal files of the Hedstrom design for use in endodontic procedures.

1.2 Classification. Hedstrom files shall be of the following classes and sizes as indicated:

Class I. - 21 millimeter, operative end.

Sizes.		
	15	50
	20	55
	25	60
	30	70
	35	80
	40	90
	45	100

Class II. - 25 millimeter, operative end.

Sizes.		
	15	50
	20	55
	25	60
	30	70
	35	80
	40	90
	45	100

2. APPLICABLE DOCUMENTS

2.1 Specifications and Standards. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

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SPECIFICATIONS

Federal

QQ-B-626	Brass, Leaded and Nonleaded. Rod, Shaped, Forgings, and Flat Products With Finished Edges (Bar and Strip).
MMM-A-250	Adhesive, Water-resistant, (For Sealing Fiberboard Boxes.)
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	Box, Fiberboard.
PPP-T-76	Tape, Pressure-sensitive Adhesive Paper, Water-resistant (For Carton Sealing).

Military

MIL-P-116	Preservation, Methods of.

STANDARDS

Federal

FED-STD-66	Steel, Chemical Composition and Hardenability.
FED-STD-151	Metal, Test Methods.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Operative end. The operative end of the Hedstrom files shall be fabricated of AISI 302 stainless steel in accordance with FED-STD-66, with a minimum tensile strength of 270,000 pounds per square inch when tested as specified in 4.4.4.

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3.1.2 Handle. Handle of the files shall be fabricated of diallyl phthalate plastic molding compound consisting essentially of diallyl phthalate, short glass fibers, titanium oxide, and coloring agent. As an alternate, handle shall be fabricated of brass conforming to copper alloy number 360, hard temper of QQ-B-626 for chemical composition and tensile strength. Brass handles shall be nickel or chromium over nickel-plated.

3.2 Style, design, and dimensions. The style, design and dimensions of the Hedstrom files shall be of the standardized system conforming to the requirements of Table I and Figure 1. The files shall be suitable for use as cutting instruments for gaining access to, clearing, enlarging, and smoothing the walls of a root canal. Hedstrom files shall be of two-piece construction, comprised of an operative end and a handle.

3.2.1 Operative end. The operative end shall consist of a conical distal tip, a uniformly tapered spiral cutting portion, and a non-tapered round shank. Spirals forming the cutting edges shall be right hand and shall extend to the tip. The length of spiralled section (B2) from free distal end shall be 15 $\pm$ 1 millimeters (mm.). The length of operative end (B3) shall be 21 $\pm$ 1 mm. for class I files and 25 $\pm$ 1 mm. for class II files. The operative end shall be securely and permanently affixed in the handle with no movement of operative end permitted.

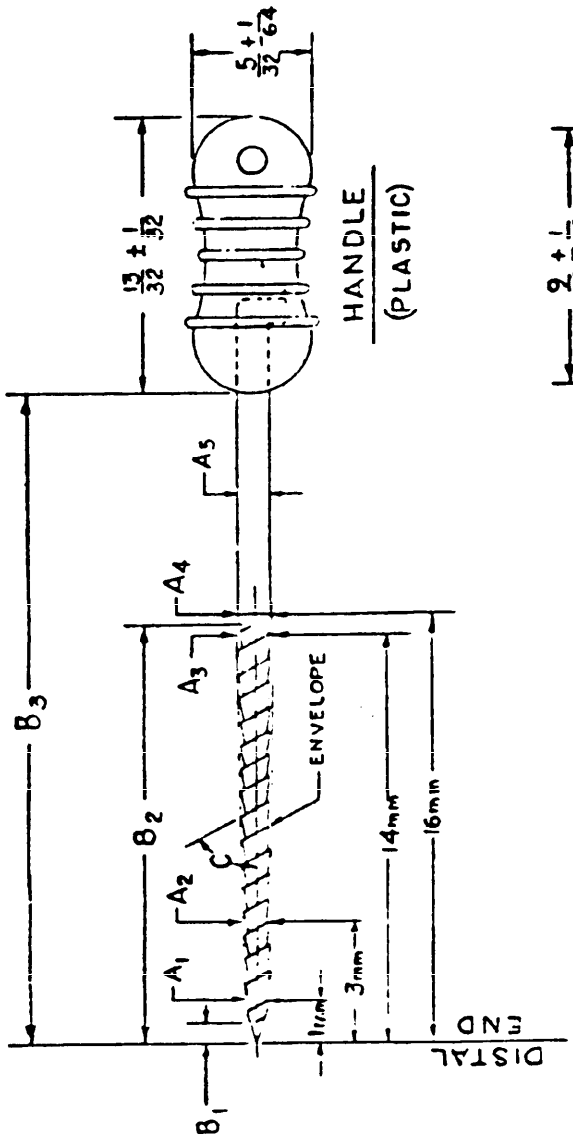
3.2.2 Handle. Handle shall be plastic or brass in accordance with style, design and dimensions shown in figure 1. Plastic handles shall have a hole as shown in figure 1 for attachment of ligature. Brass handles shall have a groove for attachment of ligature.

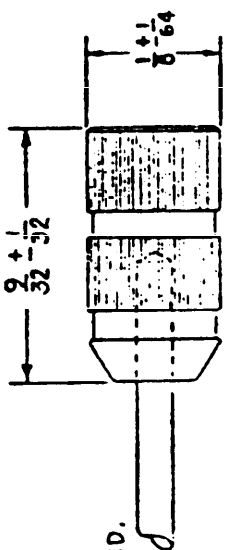
3.3 Hardness. Operative ends of files shall have a Knoop hardness number of 510 to 670 (Rockwell C48 to C57) when tested as specified in 4.4.2.

3.4 Corrosion resistance. Files shall show no corrosion, when tested as specified in 4.4.3.

3.5 Finish. Operative ends of files shall have a bright and smoothly honed finish. Handles shall have serrations or grooves for positive gripping of the handle as shown in figure 1.

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ALTERNATE HANDLE
(PLATED BRASS)

LEGEND

A_1 - DIAMETER OF ENVELOPE 1mm FROM DISTAL END.
A_2 - " " " 3mm " " " "
A_3 - " " " 14mm " " " "
A_4 - " " " 16min " " " "
A_5 - DIAMETER OF UNCUT WIRE

B_1 - LENGTH OF TIP FROM DISTAL END.
B_2 - LENGTH OF SPIRALLED PORTION FROM DISTAL END.
B_3 - LENGTH OF OPERATIVE END.

C - CUTTING EDGE ANGLE

FILE, PULP CANAL, DENTAL, HEDSTROM

FIGURE 1

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Table I. Dimensions of Hedstrom Files

Standardized size Numbers	A1 (mm.) Ref.	A2 (mm.) ± 0.02	A3 (mm.) ± 0.02	A4 (mm.) Ref.	A5 (mm.) Diameter of shank ± 0.02	Number of spirals ± 1	B1 (mm.) tip length ± 0.1	/C (degrees) Cutting edge angle ± 1 degree
15	0.15	0.21	0.43	0.45	0.80	21	0.6	78.0
20	0.20	0.26	0.48	0.50	0.80	21	0.6	77.0
25	0.25	0.31	0.53	0.55	0.80	20	0.6	76.5
30	0.30	0.36	0.58	0.60	0.80	20	0.6	76.0
35	0.35	0.41	0.63	0.65	0.80	19	0.6	75.5
40	0.40	0.46	0.68	0.70	0.80	18	0.7	75.0
45	0.45	0.51	0.73	0.75	1.00	17	0.7	74.5
50	0.50	0.56	0.78	0.80	1.00	17	0.7	74.5
55	0.55	0.61	0.83	0.85	1.00	17	0.7	74.0
60	0.60	0.66	0.88	0.90	1.20	16	0.7	73.5
70	0.70	0.76	0.98	1.00	1.20	16	0.9	73.0
80	0.80	0.86	1.08	1.10	1.20	15	0.9	72.0
90	0.90	0.96	1.18	1.20	1.40	14	0.9	70.5
100	1.00	1.06	1.28	1.30	1.40	14	0.9	70.0

Note: A1 and A4 are referenced dimensions which are difficult to measure and therefore are used only as a basis for calculating the values of A2 and A3 which are to be measured. The calculation is based on a uniform taper from A1 to A4 of 0.01875 mm. of diameter per mm. of instrument length.

3.6 Identification marking. Plastic handles shall be color-coded and the color shall be incorporated in the plastic formulation. A label or instruction sheet shall be included in each unit package giving the complete color code system as follows:

Color	Size	Color	Size
White	15, 45, 90	Blue	30, 60
Yellow	20, 50, 100	Green	35, 70
Red	25, 55	Black	40, 80

Brass handles shall have the applicable size number stamped on butt end of handle.

3.7 Cold bends. The uncut portion of the operative end of the file shall withstand a minimum of 4 bends for sizes 15 through 40 and a minimum of 2 bends for sizes 45 through 100, when tested as specified in 4.4.5.

3.8 Workmanship. Workmanship shall be first class throughout. The files shall be free from defects which detract from their appearance or may impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the component or material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the Item Identification, the name of the component or material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables II and IV (see 4.2.1 and 4.3), and table III (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table II. Unit of product for examination purposes shall be the unit of issue (6 files).

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Table II. Sampling for Examination

Type defect	Inspection level	AQL (Percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table III. Unit of product for test purposes shall be as indicated in table III.

Table III. Tests

Component	Characteristic	Requirement	Test procedure	Unit of Product	Inspection level	AQL (percent defective)
Operative ends	Tensile strength	3.1.1	4.4.4	Lot of steel stock	*	---
	Bend test	3.7	4.4.5	Individual file	S-2	**
	Hardness	3.3	4.4.2		S-1	**
Entire Instrument	Corrosion-resistance	3.4	4.4.3		S-2	**

* Average of a minimum of three specimens from each lot of steel stock.
** Acceptance number shall be zero for all sample sizes.

4.2.3 Raw materials. Sufficient material from each lot of metal used in the construction of the files shall be selected for chemical analysis.

4.2.4 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

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4.3 Examination. The files shall be examined for defects including, but not limited to, those listed in table IV.

Table IV. Classification of Defects

Categories	Defects
Major A:	
101	Class and size not as specified.
102	Style and design not as specified.
103	Hole (in plastic handle) or groove (in brass handle) missing.
104	Plastic handles not color coded.
105	Label or instruction sheet giving color code system not included in unit package (plastic handles only).
106	Identification marking missing on brass handle.
107	Files not of two-piece construction.
108	Operative ends do not have a bright smoothly-honed finish.
109	Cutting edges not sharp.
110	Operative ends not firmly affixed in the handles.
111	Laminations or cracks present.
112	Files bent in package.
113	Operative end bent, deformed, or otherwise damaged.
Major B:	
151	Number of spirals not as indicated for class and size number.
152	Dimensions of operative end not as specified.
Minor A:	
201	Finish of handles not as specified.
202	Packaging and packing not as specified.
Minor B:	
251	Dimensions of handles not as specified.

4.4 Tests.

4.4.1 Chemical analysis of the metals. In the absence of certificates of quality for the raw material, the samples selected in accordance with 4.2.3 shall be analyzed in conformance with methods 111 or 112 of FED-STD-151. In case of dispute, method 111 shall be used for referee analysis.

4.4.2 Hardness test. The Knoop hardness number shall be determined on a Tukon tester under a 500-gram load, using a rhombus-shaped diamond indenting tool with a longitudinal angle of 172.5 degrees and a transverse angle of 130 degrees.

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4.4.3 Corrosion-resistance. Before testing, the files shall be scrubbed with soap and water, rinsed thoroughly in distilled water, and dried. After cleaning, the files shall be boiled in distilled water in a glass beaker for 30 minutes and allowed to remain submerged for 24 hours.

4.4.4 Tension test. Tension test shall be performed in a suitable standard tension tester on an 8-inch sample of steel stock used to produce the files.

4.4.5 Bend test. The files shall be subjected to a bending test on a bending machine at a uniform rate of 90 degrees per second, alternating right and left. Each 90 degree movement in either direction shall be counted as one bend. Distance between jaws shall be 3/32 inch.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Unit of issue. One package, containing 6 files, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities The number of units of issue indicated in Table V shall be packaged in each unit, intermediate, and exterior container as applicable, for the required level of protection specified in the procurement document.

Table V. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	24 units	576 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be packaged in a suitable plastic box that will allow the files to be seen. Cover of box shall be a slipover or hinged cover. Box shall be designed with six cavities, so that six files can be packaged without cushioning. The cavities shall be molded, so as to hold the files in place in such a manner that no free movement is present to damage the operative ends of the files. The cover, when placed or closed shall be flush on the files. Cover shall be secured to prevent accidental opening. Each unit shall be further packaged in accordance with MIL-P-116, method IA-8.

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5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flap with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class and size required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Supersession data. This specification includes the requirements of Purchase Description No. 1, dated 21 February 1968.

6.3 Limitations. This specification does not cover all classes or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6520-1293

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R295

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION

MIL-F-36782

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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CAUTION

NOTICE TO BIDDERS OFFERORS

DO NOT CONDITION OR BASE YOUR BID OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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